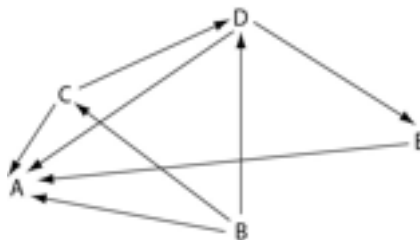


Read each question completely before answering. Each question is worth 2 points except as noted.

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

- In ecosystems, why is “cycling” used to describe the transfer of chemical elements and nutrients, while “flow” is used for the transfer of energy?
 - Chemical elements are recycled within ecosystems, but energy flows through and out of ecosystems.**
 - Chemical elements are cycled into ecosystems from other ecosystems, but energy constantly flows within the ecosystem.
 - Both chemical elements and energy are recycled but can flow to other ecosystems.
 - Both chemical elements and energy flow within an ecosystem but can cycle to other ecosystems.
 - Energy flows from one ecosystem to other ecosystems, but chemical elements constantly cycle within an ecosystem.
- Plants produce water vapor during photosynthesis, which goes from the plant to the atmosphere. The water vapor in the atmosphere eventually falls back to the earth as rain. This process would be part of
 - the nutrient cycle.
 - the water cycle.**
 - plant metabolism.
 - energy flow.
 - the energy cycle.
 - the entire ecosystem.
- Consider the food chain grass → grasshopper → mouse → snake → hawk. How much of the chemical energy fixed by photosynthesis of the grass (100%) is available to the hawk?
 - 0.01%**
 - 0.1%
 - 1%
 - 10%
 - 60%
- Based on the hypothetical food web below, one letter represents an organism that could be both a secondary and tertiary consumer? The arrows point from lower to higher trophic levels. More than one answer is possible. Circle only one answer.



- A**
- B
- C
- D
- E

5. The land on the south side of the Himalayan mountain range (below) receives relatively high levels of rainfall and has fertile farmlands. The land north of the mountain range is hot and dry and can't be farmed. Which of the following is true.

- A. The south side of the Himalayan mountain range is in a rain shadow.
- B. The cool, moist Indian Ocean air heats up as it rises, releasing its precipitation as it passes the tops of the mountains, and this warm, now dry air cools as it descends on the leeward side of the range.
- C. The desert in southern China warms the north side of the mountain range, causing it to dry out.
- D. The warm, moist air is too dense to rise over the height of Mount Everest and the other mountains in this range so the moisture remains on the south side of the range.
- E. **Warm, moist air from the Indian Ocean blows north, rises and cools, releasing precipitation as it moves up the south side of the range, and this cool, now dry air mass heats up as it descends on the north side of the range.**

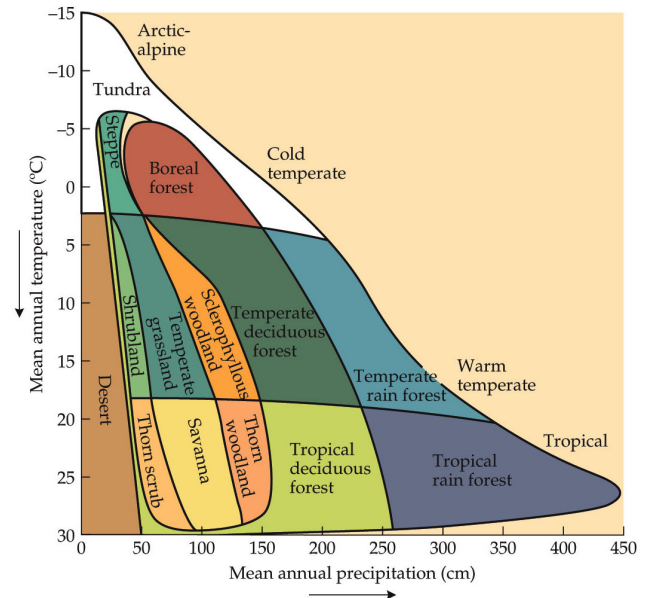


6. A tertiary consumer has high abundance. In a trophic cascade, which of the following is probably *not* true.
- A. **The primary producer will have high abundance.**
 - B. The decomposer level will have low abundance.
 - C. The secondary consumer will have low abundance.
 - D. The decomposer level will have high abundance.
 - E. The primary consumer will have high abundance.
 - F. The primary producer will have low abundance.
7. In summer 2012, the Midwest experienced one of the worst droughts in recorded history. Which of the following statements is the best explanation of how this drought affected the Gulf of Mexico dead zone during summer 2012?
- A. The dead zone will be smaller because farmers, anticipating the drought, would not have added fertilizer to the crop fields the spring before the drought.
 - B. **The dead zone will be smaller because the Mississippi River will carry less water so fewer nutrients will be carried to the Gulf of Mexico.**
 - C. The dead zone will be larger because the reduced volume of water will concentrate the nutrients that enter the Gulf of Mexico.
 - D. The dead zone will be similar to previous years because the drought is not likely to affect the amount of water that flows out through the Mississippi River.
 - E. The dead zone will be smaller because less freshwater will enter the Gulf of Mexico, allowing more saltwater fishes to live around the mouth of the river.
-

8. Which of these species is most likely a keystone species in Alaska?
- A. Sea urchins.
 - B. Kelp.
 - C. Killer whales.
 - D. Humans.
 - E. Sea otters.**
9. A fish ate shrimp. The shrimp ate algae (which are photosynthetic). The algae is a _____. The fish is a _____.
- A. primary consumer; primary producer.
 - B. secondary consumer; primary producer.
 - C. primary producer; secondary consumer.**
 - D. primary consumer; secondary consumer.
 - E. secondary consumer; primary consumer.
 - F. primary producer; primary consumer.
10. Considering your knowledge of the dead zone and the causes of microscopic algae blooms, the Gulf of Mexico ecosystem is most likely to be limited by which of the following nutrients?
- A. CO₂
 - B. nitrogen**
 - C. oxygen
 - D. hydrogen
 - E. carbon
11. The dead zone in the Gulf of Mexico and other oceans around the globe occurs when
- A. an excess of nutrients causes an algal bloom. The algae consume all of the oxygen in the water for photosynthesis, so other organisms can no longer live there.
 - B. an excess of nutrients causes an algal bloom, followed by death of the algae. Decomposition of the algae uses up the oxygen in the water, so other organisms can no longer live there.**
 - C. eutrophication causes an absence of nutrients in the zone.
 - D. global climate change causes the water to become too warm to sustain life.
 - E. commercial fishers over-harvest shrimp, crab, fishes and other sea animals.
12. On average, the amount of secondary consumer biomass is about _____ less than the amount of primary producer biomass.
- A. 10×
 - B. 20×
 - C. 50×
 - D. 80×
 - E. 100×**
-

13. The Ozark region of Missouri lies within the temperate deciduous forest ecosystem. If, within the next 25 years, global climate change raises the mean annual temperature from 14°C to 18°C and lowers the mean annual precipitation from 160 cm to 130 cm, which of the following is most likely?

- A. The ecosystem will become more like temperate grassland.
- B. The ecosystem will become more like boreal forest.
- C. The ecosystem will become more like temperate rain forest.
- D. The ecosystem will become more like sclerophyllous or thorn woodland.**
- E. The ecosystem will remain temperate deciduous ecosystem.



14. The energy stored by producers can be passed through an ecosystem along a(n) _____, a series of interactions in which organisms transfer energy by eating and being eaten.
- A. trophic cascade
 - B. food web**
 - C. nutrient cycle
 - D. ecosystem
 - E. energy cycle
 - F. predator-prey chain
15. If humans stopped fishing along the coast of Alaska, what will probably happen to the kelp forest and why?
- A. The kelp abundance will increase because the sea urchin abundance will probably decrease.**
 - B. The kelp abundance will remain about the same. The kelp are not part of this community, so the kelp would not be affected.
 - C. The kelp abundance will decrease. Sea otters eat urchins and urchins eat kelp. Because there were fewer otters to eat urchins, the number of urchins increased. The urchins therefore ate more kelp so the kelp abundance decreased.
 - D. The kelp abundance will decrease because more fish will be present to eat the kelp.
 - E. The kelp abundance will increase because of the symbiotic relationship between kelp and fishes. Fishes are necessary for kelp to reproduce.

16. The Mediterranean coast and coastal California have similar types of ecosystems. This is because
- A. they are both wine-growing regions.
 - B. they have similar types of organisms.
 - C. they have similar types of communities.
 - D. they have similar climates.**
 - E. they are at similar latitudes.

Circle True or False. Read each statement completely before answering.

- 17. **TRUE** False In addition to facilitation, random effects also influence which species are present in a mature community.
 - 18. True **FALSE** Nutrients are lost from an ecosystem when an organism dies.
 - 19. True **FALSE** Desert ecosystems vary more by precipitation than by temperature.
 - 20. **TRUE** False Energy that enters into an ecosystem is eventually lost as metabolic heat.
 - 21. **TRUE** False The food web of any ecosystem depends on live producers and decomposers.
 - 22. **TRUE** False After a volcanic eruption like Mt. St. Helens, both primary and secondary succession can occur.
 - 23. True **FALSE** Net primary production represents the amount of energy that will be available to secondary consumers in an ecosystem.
 - 24. **TRUE** False Deserts at 30° latitude are dry because the moisture was left at the equator.
 - 25. True **FALSE** A field containing 30,000 kg of plant biomass will support about 300 kg of tertiary consumer biomass.
 - 26. True **FALSE** On average, about 10% of nutrients are transferred from primary producers to primary consumers.
 - 27. (5 points) Evidence suggests that average global temperatures have been much higher in the geological past compared to the current average global temperature. If so, then explain why we should be concerned about global climate change now?
-

Because the rate of climate change is rapid. Organisms may not have the time to adapt to changing climate.

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Algae bloom	Gross primary production	Primary production
Biomass	Herbivore	Primary succession
Blanket effect	Inhibition	Secondary consumer
Decomposer	Net primary production	Secondary succession
Density	Mean annual precipitation	Solar radiation
Energy flow	Mean annual temperature	Specialist
Eutrophication	Nitrification	Tertiary consumer
Facilitation	Nutrient recycling	Tertiary succession
Generalist	Primary consumer	Tolerance
Greenhouse effect	Primary producer	Trophic cascade

28. **Generalist** Type of species most commonly found in early stage of succession.
29. **Net primary production** Amount of sun's energy transferred to primary consumers as organic matter.
30. **Solar radiation** The single most important factor that determines global climate patterns.
31. **Decomposer** Group of organisms that get energy from dead organisms.
32. **Greenhouse effect** Keeps Earth's atmosphere at a suitable temperature for life.
33. **Facilitation** Intermediate successional species change the environment, making easier for late successional species to colonize.
34. **Biomass** The mass of living organisms in an area.
35. **Primary producer** Responsible for gross primary production.
36. **Eutrophication** Excess nutrients in an ecosystem.
37. **Secondary succession** Following disturbance, begins from soil.
38. **Trophic cascade** Upper trophic level species causes alternating abundance levels of lower trophic level species.
39. Name of your favorite movie.
-

A scientist gave 30 individuals a choice of five habitats. She predicted that more individuals would be found in chambers 2 and 3 compared to chambers 1, 4, or 5. Her results are shown in the figure. Her calculated χ^2 was 9.33.

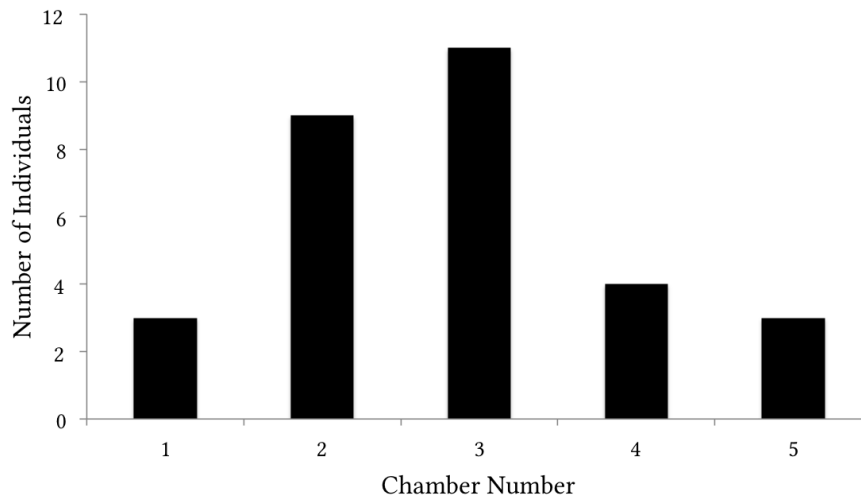


Table 1. Critical values of χ^2 .

df	α		
	0.10	0.05	0.01
1	2.71	3.84	6.63
2	4.61	5.99	9.21
3	6.25	7.81	11.34
4	7.78	9.49	13.28
5	9.24	11.07	15.09
6	10.64	12.59	16.81
7	12.02	14.07	18.48
8	13.36	15.51	20.09
9	14.68	16.92	21.67
10	15.99	18.31	23.21
11	17.28	19.68	24.72
12	18.55	21.03	26.22
13	19.81	22.36	27.69
14	21.06	23.68	29.14
15	22.31	25.00	30.58
16	23.54	26.30	32.00
17	24.77	27.59	33.41
18	25.99	28.87	34.81
19	27.20	30.14	36.19
20	28.41	31.41	37.57
21	29.62	32.67	38.93
22	30.81	33.92	40.29
23	32.01	35.17	41.64
24	33.20	36.42	42.98
25	34.38	37.65	44.31
26	35.56	38.89	45.64
27	36.74	40.11	46.96
28	37.92	41.34	48.28
29	39.09	42.56	49.59
30	40.26	43.77	50.89
31	41.42	44.99	52.19
32	42.58	46.19	53.49
33	43.75	47.40	54.78
34	44.90	48.60	56.06
35	46.06	49.80	57.34

40. (2 points) How many degrees of freedom did she have? **4**
41. (2 points) What critical value of χ^2 did she use? **9.49**
42. (5 points) Did she accept or reject her hypothesis? Explain why.

Reject her hypothesis. Calculated χ^2 is less than the critical value. If calculated is less than critical, then she accepts (or fails to reject) the null hypothesis and rejects her research hypothesis.

43. (5 points) Explain why climate change is a better term than global warming to describe the effects of warming due to the accumulation of greenhouse gases.

Because climate is composed of mean annual temperature and mean annual precipitation. Warming temperatures will affect precipitation patterns.

44. (5 points) NPP is always lower than GPP. What happens to the difference?

Plant metabolism.

45. Select whether you want **2 points** or **5 points** added to your score for this exam. Before you decide, there is a small catch: if more than 10% of the class selects 5 points, then no one gets the extra points. Only I will see your choice.

A. 2 points. B. 5 points.

Count the day as lost that you do not learn something new. Stay curious, do good things, and have a great summer!
